Contents

Building Frontend for a Home Listings Admin App using ReactJS **2**

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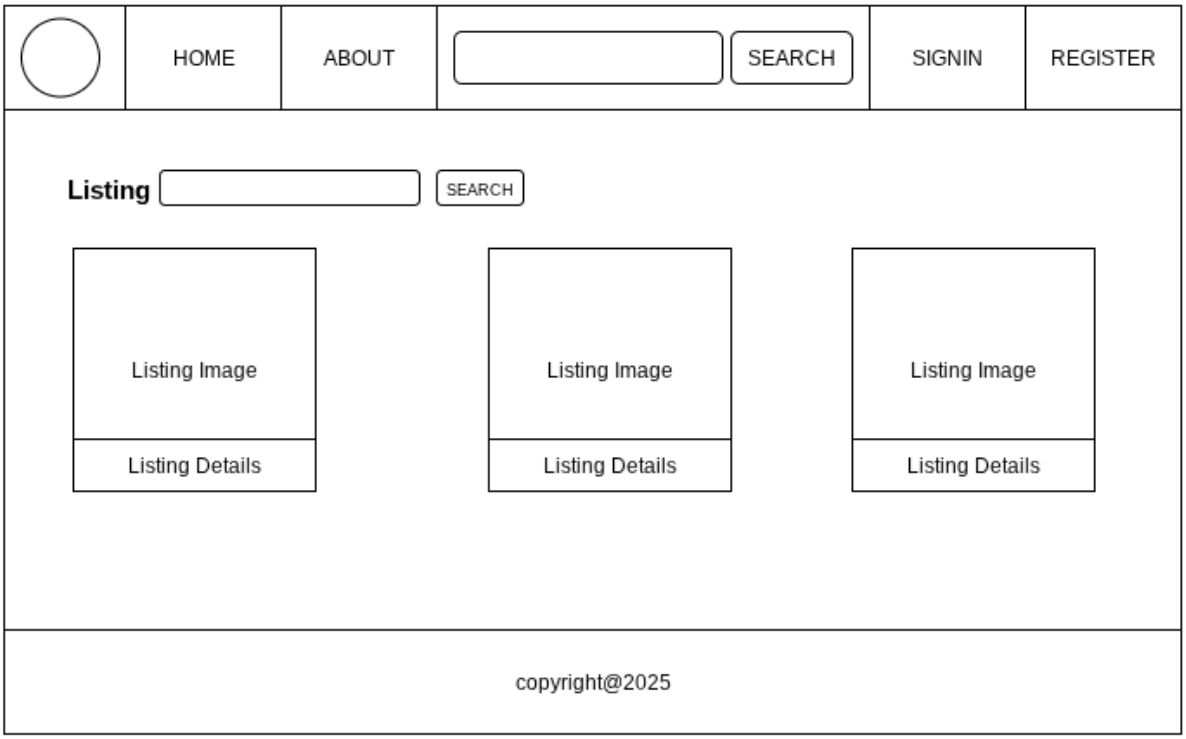
Building Frontend for a Home Listings Admin App using ReactJS

Before building any frontend app from scratch, we need to do a preparatory step of designing the frontend. While we have many design steps that we follow, we will see a brief of those here:

- Starting with an idea: Home `Listings Admin` App For a Home Listings app, we can have a seperate admin interface & than for a normal Customer. Here we will focus on building an Admin Interface.
- Design the Low Level Design: Taking inspiration from other works & deciding on the design, we iterate many time to reach at the desired UI. We decide on how the elements are placed & tentative structure of each element.
  This can be done on pen & paper or any software online. We will consider for our app, as of now 3 pages.
- Home Page
- Dashboard
- Listing Edit

HomePage



Dashboard

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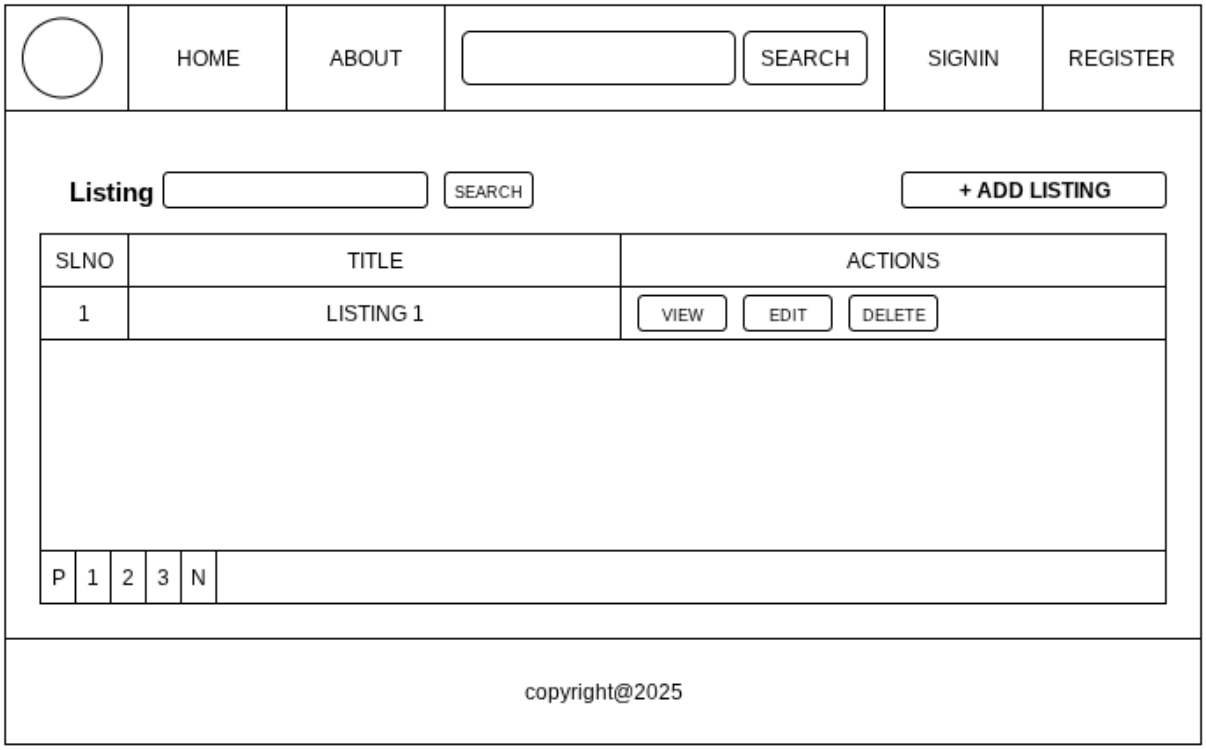
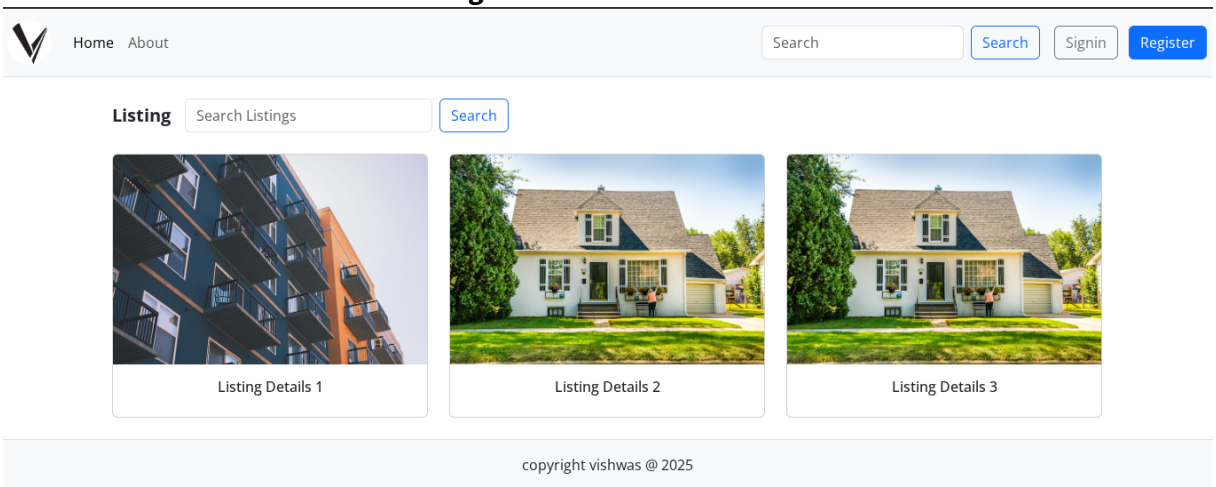


After this we work on typography (fonts) and Color Theory - primary, secondary & tertiary colors for our website, (70-20-10 rule etc..). We come up with a highlevel design

HomePage



Dashboard

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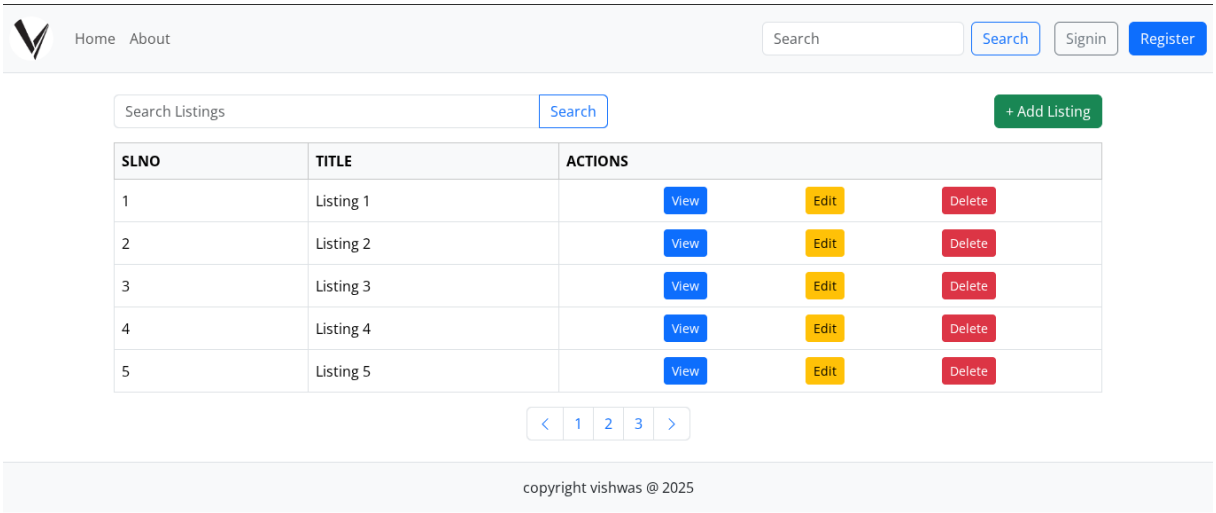
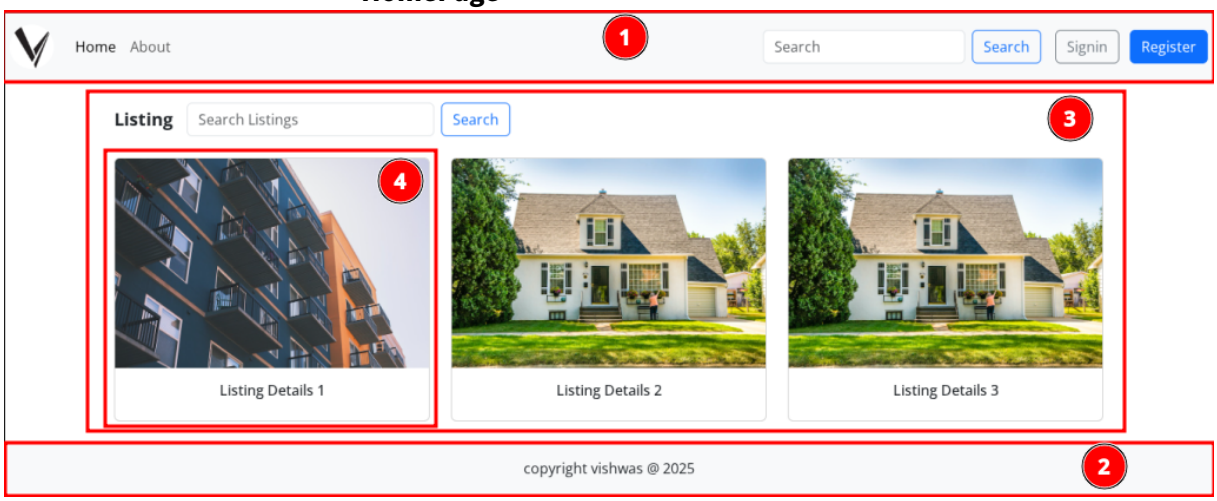


Once we design the pages as above we can break them into components as below

HomePage



Dashboard

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We can drilldown into more detail components. With the above pictures we can create

- Page Components
  - HomePage
  - About
  - Dashboard and many more
- Individual Sub-Components
  - Layout
  - Header/AppNavBar (1)
  - ListingCard (3)
  - AppFooter (2)
  - ListingsTable (4)
  - ListingRow (5) and many more

We want our data to be seperate from our UI so we keep a seperate services layer which will be responsible for providing data for us. The project structure will be similar to as below

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```
my-app/
├── public/
│   ├── index.html
│   └── favicon.ico
│
├── src/
│   ├── assets/                # Images, icons, styles, fonts
│   │   ├── images/
│   │   └── styles/
│
│   ├── components/            # Reusable UI components
│   │   ├── layout/
│   │   │   ├── Layout.jsx
│   │   │   ├── Header.jsx      # (AppNavBar)
│   │   │   ├── Header.css
│   │   │   └── Footer.jsx      # (AppFooter)
│   │   │   ├── Footer.css
│   │   │
│   │   ├── listings/
│   │   │   ├── ListingsTable.jsx
│   │   │   ├── ListingsTable.css
│   │   │   ├── ListingRow.jsx
│   │   │   ├── ListingRow.css
│   │   │
│   │   ├── common/            # Buttons, inputs, dialogs, etc.
│   │   │   ├── CustomButton.jsx
│   │   │   └── SearchBox.jsx
│   │   │
│   │   └── index.js           # Optional barrel export
│
│   ├── pages/                 # Full page views
│   │   ├── HomePage.jsx
│   │   ├── AboutPage.jsx
│   │   ├── DashboardPage.jsx
│
│   ├── services/              # API & business logic
│   │   ├── api/
│   │   │   ├── listingService.js  # CRUD listings
│   │   ├── httpClient.js          # axios/fetch wrapper (Future)
│   │   └── storageService.js      # localStorage/session helpers (Future)
│
│   ├── context/               # React Context API (Future)
│   │   ├── AuthContext.jsx
│   │   └── ListingsContext.jsx
│
│   ├── hooks/                 # Custom React hooks(Future)
│   │   ├── useAuth.js
│   │   └── useFetch.js
│
│   ├── routes/                # Routing layer (Future)
│   │   └── AppRoutes.jsx
│
│   ├── App.jsx                # Root component with Layout
│   ├── index.js               # Entry point
│   └── setupTests.js          # Jest/RTL tests
│
└── package.json
```

Now we will start with building our project. We can use local environment or online environment like Github Codespaces.

Let us use codespaces & hence we will

- create a git repo for our listingsapp.

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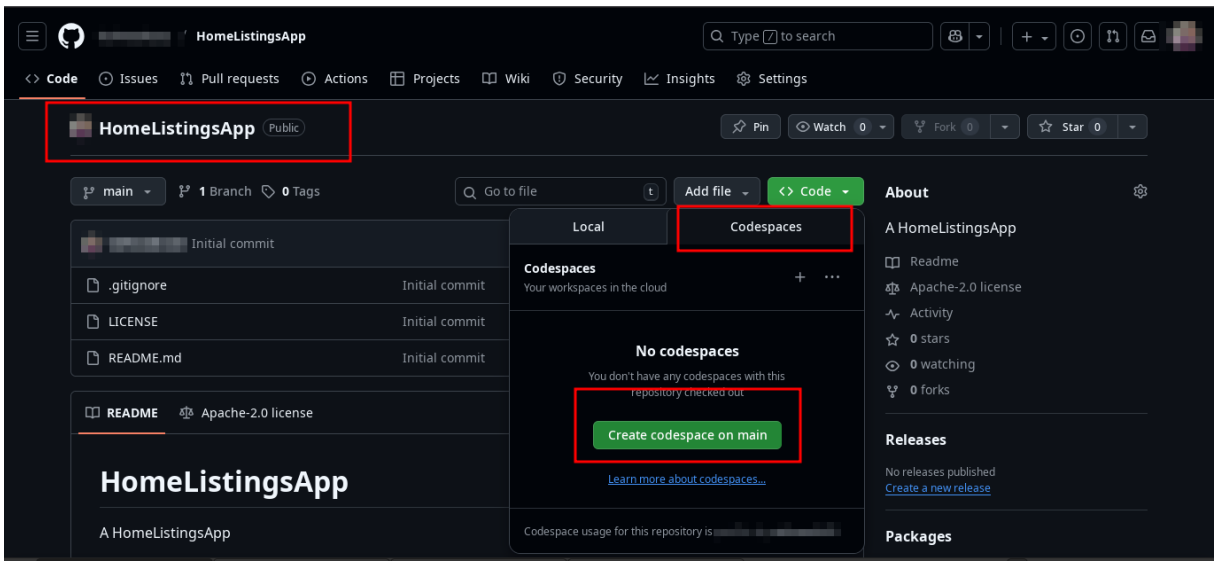
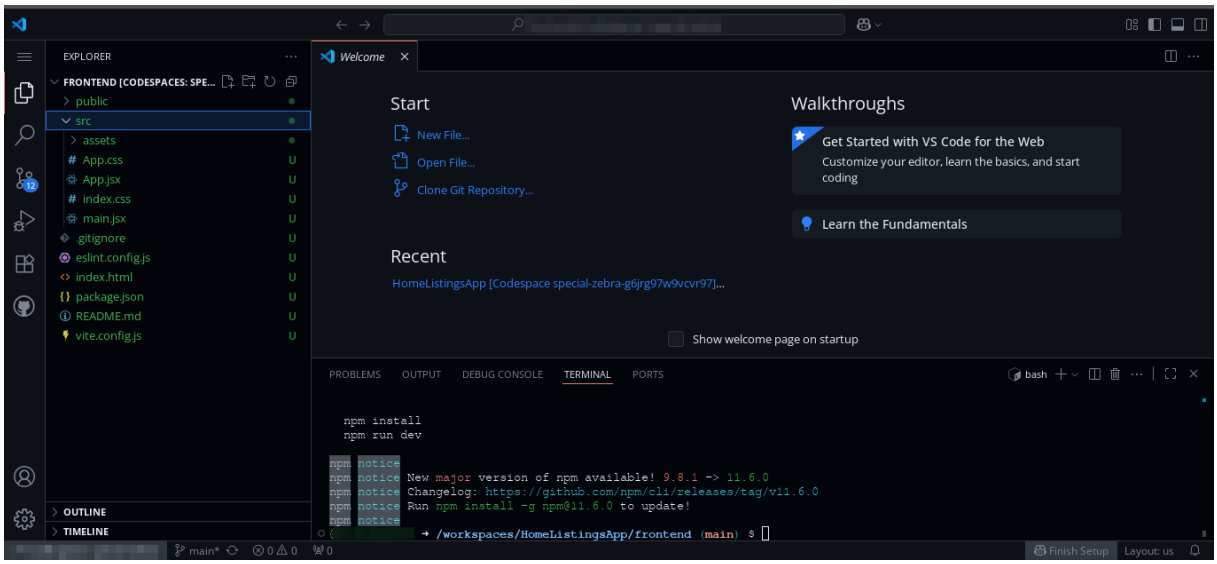


- We will use the below command to create a react app using vite in current folder

```
npm create vite@latest .
```

follow the oncreeen instructions to select `react`, `javascript`



Next we have to install the dependencies

```
npm install
```

We will start the development server

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```
npm run dev
```

We will start our development by

- delete `react.svg` in assets
- clear contents of `App.css` & `App.jsx`, `index.css`
- We will create the necessary folders of `components`, `pages`, `services`

```
mkdir -p src/assets/images src/assets/styles
mkdir -p src/services/api
mkdir -p src/components/layout src/components/listings
↪  src/components/common
mkdir -p src/pages
```

- next we will add the bootstrap to our project

```
npm install bootstrap
```

import once in `src/index.jsx`

```
import 'bootstrap/dist/css/bootstrap.min.css';
```

just use Bootstrap classes directly in JSX. Also instead of using raw divs everywhere, we can use react-bootstrap which gives us sanitized components.

```
npm install react-bootstrap
```

Instead of writing raw Bootstrap classes, you can use React components.

- create `AppNavbar`, `Footer`, `Layout` Components.
  In General to build a component we do

1. Export the component
2. Define the function
3. Add markup

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Example

```
export default function Profile() {
  return (
    <img
      src="https://i.imgur.com/MK3eW3Am.jpg"
      alt="Katherine Johnson"
    />
  )
}
```

4. Using a component we can nest it inside other components

AppNavBar

```
import { Navbar, Nav, Container, Button, Form, FormControl } from
↪  'react-bootstrap';

function AppNavBar() {
  return (
    <Navbar bg="light" expand="lg" className="border-bottom">
      <Container>
        <Navbar.Brand href="#">MyApp</Navbar.Brand>
        <Navbar.Toggle aria-controls="basic-navbar-nav" />
        <Navbar.Collapse id="basic-navbar-nav">
          <Nav className="me-auto">
            <Nav.Link href="/">Home</Nav.Link>
            <Nav.Link href="/about">About</Nav.Link>
          </Nav>
          <Form className="d-flex me-2">
            <FormControl type="search" placeholder="Search"
            ↪  className="me-2" />
            <Button variant="outline-primary">Search</Button>
          </Form>
          <Button variant="outline-secondary"
          ↪  className="me-2">Signin</Button>
          <Button variant="primary">Register</Button>
        </Navbar.Collapse>
      </Container>
```


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```
    </Navbar>
  );
}

export default AppNavBar;
```

Footer.jsx

```
import { Container } from "react-bootstrap";

export default function Footer() {
  return (
    <footer className="bg-light border-top py-3 mt-auto">
      <Container className="text-center">
        <small className="text-muted">
          © vishwas @ 2025
        </small>
      </Container>
    </footer>
  );
}
```

Layout.jsx

```
import AppNavBar from "./AppNavBar";
import Footer from "./Footer";
import { Container } from "react-bootstrap";

export default function Layout({ children }) {
  return (
    <div className="d-flex flex-column min-vh-100">
      <AppNavBar />
      <Container className="flex-grow-1 my-4">
        {children}
      </Container>
      <Footer />
    </div>
  );
}
```

HomePage

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```
export default function HomePage() {
  return (
    <div>
      <h1>Welcome to MyApp</h1>
      <p>This is the homepage content.</p>
    </div>
  );
}
```

We will use these in `App.jsx`

```
import Layout from "./components/layout/Layout";
import HomePage from "./pages/HomePage";
function App() {
  return (
    <Layout>
      <HomePage />
    </Layout>
  );
}

export default App;
```

Output

We will start with a list of homeListings in HomePage.

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```
let listings = [
  { id: 1, title: "Cozy Apartment in City Center", image:
↪  "public/images/house.jpg" },
  { id: 2, title: "Spacious Villa with Garden", image:
↪  "public/images/house.jpg" },
  { id: 3, title: "Modern Studio near Metro", image: "public/images/house.jpg"
↪  },
  { id: 4, title: "Luxury Penthouse with Pool", image:
↪  "public/images/house.jpg" },
  { id: 5, title: "Family Home in Suburbs", image: "public/images/house.jpg" },
];
```

We will create a card component to display on home page **ListingCard.jsx**

```
import { Card, Button } from "react-bootstrap";

export default function ListingCard({ listing }) {
  return (
    <Card className="shadow-sm h-100">
      <Card.Img
        variant="top"
        src={listing.image}
        alt={listing.title}
        style={{ height: "200px", objectFit: "cover" }}
      />
      <Card.Body className="d-flex flex-column">
        <Card.Title className="fs-6">{listing.title}</Card.Title>
        Listing Details
      </Card.Body>
    </Card>
  );
}
```

Use that in HomePage

```
import { Container, Row, Col } from "react-bootstrap";
import ListingCard from "../components/listings/ListingCard";

let listings = [
  { id: 1, title: "Cozy Apartment in City Center", image:
↪  "public/images/house.jpg" },
```


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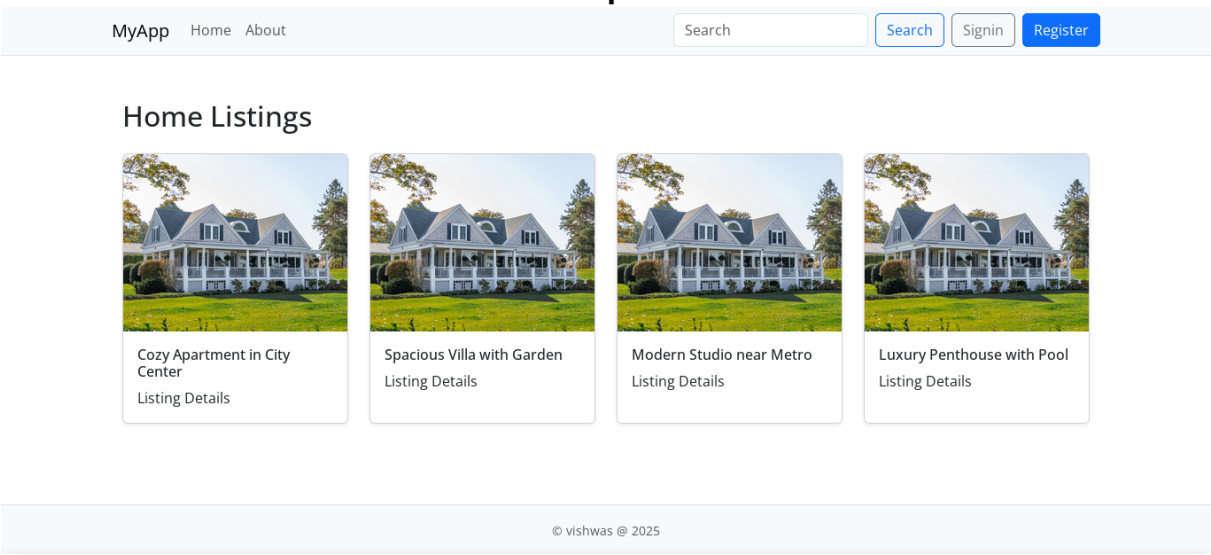
```
  { id: 2, title: "Spacious Villa with Garden", image:
↪   "public/images/house.jpg" },
  { id: 3, title: "Modern Studio near Metro", image: "public/images/house.jpg"
↪   },
  { id: 4, title: "Luxury Penthouse with Pool", image:
↪   "public/images/house.jpg" },
];

export default function HomePage() {
  return (
    <Container className="my-4">
      <h2 className="mb-4">Home Listings</h2>
      <Row xs={1} sm={2} md={3} lg={4} className="g-4">
        {listings.map((listing) => (
          <Col key={listing.id}>
            <ListingCard
              listing={listing}
            />
          </Col>
        ))}
      </Row>
    </Container>
  );
}
```

Output



For keeping the data seperate we will populate a sample data and simulate api in services.

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- Move listings data to services folder data.json
- import data in HomePage In HomePage

```
import { Col, Container, Row } from "react-bootstrap"
import ListingCard from "../components/listings/ListingCard"
import listings from "../services/data"
```

Services/data.js

```
const listings = [
    {id: 1, title: "Listing 1", image: "public/images/house.jpg"},
    {id: 2, title: "Listing 2", image: "public/images/house.jpg"},
    {id: 3, title: "Listing 3", image: "public/images/house.jpg"}

]

export default listings;
```

In the card, let us display brief description we will have know `more...` for accessing more information.

In the card component create a link with link text know `more...` define this as a variable

```
const linkText = 'know more...'
```

In the element

```
<a
  href="#"
  className="text-blue-600 hover:text-blue-800 font-medium
↪  transition-colors duration-200"
>
  {linkText}
</a>
```


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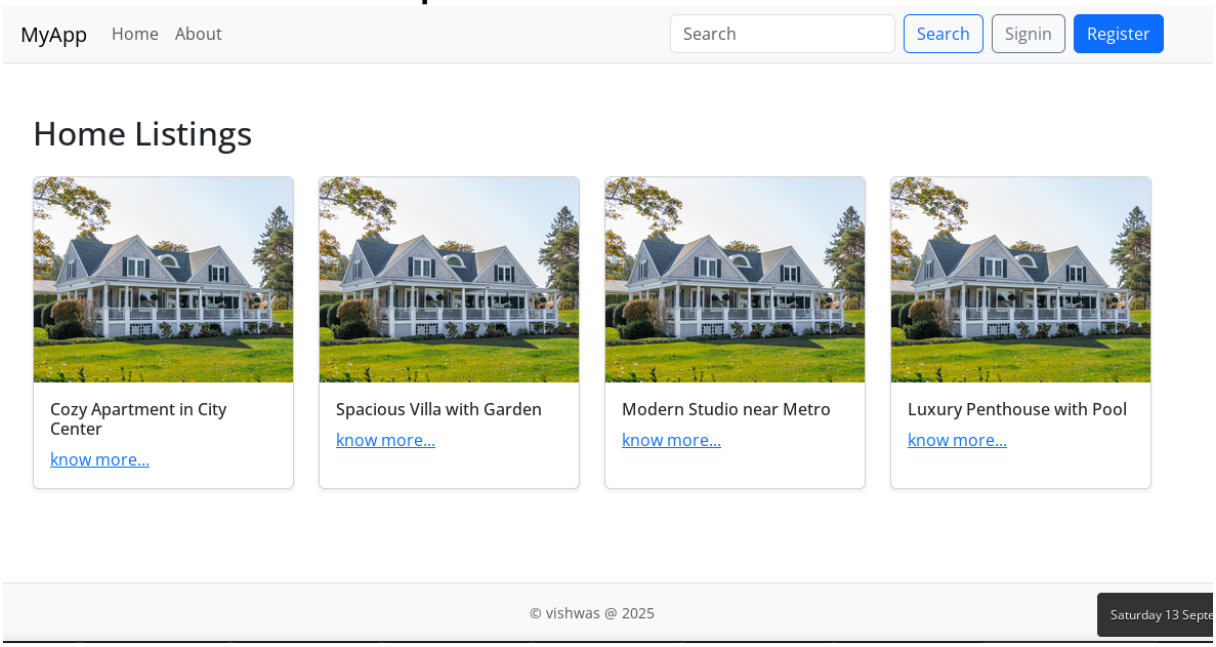
Output:



Let us make a change to our house image, when clicked should display a modal with bigger image. This requires 2 things

- Event (To trigger `click`)
- State (To keep information about whether to keep modal open or close it.)

To understand about Modal, react-bootstrap-modal

1. Attach Event Handler to image

```
<Card.Img
  variant="top"
  src={listing.image}
  alt={listing.title}
  style={{ height: "200px", objectFit: "cover" }}
  // Attach the event handler to the image.
  onClick={handleShow}
/>
```

2. Define the event handler
   Inside the card component define the handler.

```
const handleShow = () => console.log('Event called...');
```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3. Defining the Modal Component
   We can define it as a seperate component. This modal component uses a data(boolean value) to handle its open & close. A function to control its value will help us toggle it.

```
let showModal = false;
const handleClose = ()=>{
    showModal = false;
}
```

Understanding State Management

State is a way for a component to remember information. When a component's state changes, React automatically re-renders the component to show the updated information. Think of it as a component's memory.

For our modal, we need to "remember" whether the modal should be open or closed. We'll use a state variable for this, called showModal, and a function to update it, called setShowModal.

We'll use the useState hook from React to manage this state. This hook gives us a state variable and a function to update it.

```
  // State variable to track the modal's visibility.
  const [showModal, setShowModal] = useState(false);
```

The event to trigger should come from card component as props, so we can lift our state to parent component i.e `ListingCard`

1. Import use state in listing card

```
import { useState } from "react";
```

2. define listing modal component

```
import React from "react";
import { Card, Button, Modal } from "react-bootstrap";

export default function ListingImageModal({ listing, showModal,
↪   handleClose}){
    return (
        <Modal show={showModal} onHide={handleClose} centered size="lg">
        <Modal.Header closeButton>
```


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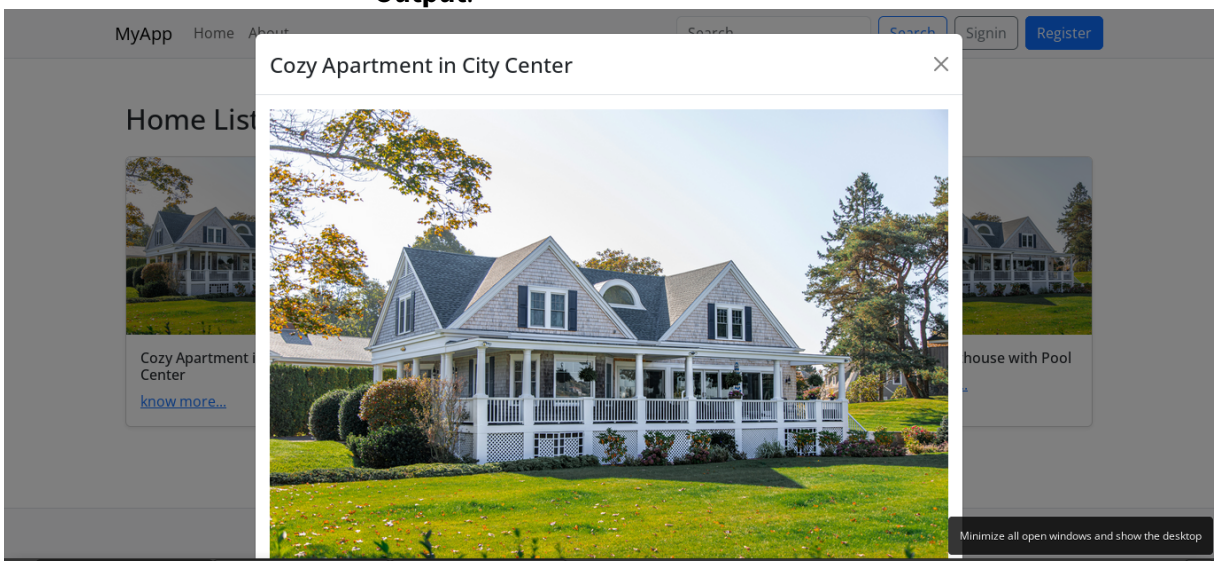
```
          <Modal.Title>{listing.title}</Modal.Title>
        </Modal.Header>
        <Modal.Body>
          <img
            src={listing.image}
            alt={listing.title}
            className="img-fluid"
            style={{ width: '100%' }}
          />
        </Modal.Body>
      </Modal>
    )
}
```

3. Call the ListingImageModal in ListingCard

```
<ListingImageModal
  listing={listing}
  showModal = {showModal}
  handleClose = {handleClose}
/>
```

We want to make users know that they can click on image so we will add on hover property to display cursor.

Output:



Now we will add routing to our pages.

Routing We have different modes of routing in React

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1. Framework Mode
2. Data Mode
3. Declarative Mode

- Framework mode incorporates both
- we will use datamode & declarative mode here.
- add the package `react-router`

```
npm i react-router
```

- import in `main.jsx`

```
import { BrowserRouter } from 'react-router'
```

- wrap the App component with browser router

```
createRoot(document.getElementById('root')).render(
  <StrictMode>
    <BrowserRouter>
      <App />
    </BrowserRouter>
  </StrictMode>,
)
```

- Create a folder named `routes`
- Define the routes in `routes/AppRoutes.jsx` **AppRoutes.jsx**

```
import { Routes, Route } from "react-router" // Don't forget the -dom part!
import HomePage from "../pages/HomePage"
import AboutPage from "../pages/AboutPage" // Import the AboutPage
↪  component

const AppRoutes = () => {
    return (
        <Routes>
            <Route path="/" element={<HomePage />} />
            <Route path="/about" element={<AboutPage />} />
        </Routes>
    )
}

export default AppRoutes // Export the component
```


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- call the AppRoutes component inside the layout

```
<Layout>
  <AppRoutes />
</Layout>
```

- Create About Page

```
import React from "react";
import { Container } from "react-bootstrap";

export default function AboutPage(){
    return (
        <Container className="my-4">
        <h1>Welcome to About</h1>
        <p>
            This is a Home Listings App.
        </p>
        </Container>
    )
}
```

- Add Active style in the AppNavbar component

```
const activeStyle = {
  color: '#007bff', // Bootstrap's primary blue
  fontWeight: 'bold',
};
```

- Modify the AppNavBar Nav Link

```
<Nav.Link as={NavLink} to="/" end style={({ isActive }) => (isActive ?
↪  activeStyle : undefined)}>Home</Nav.Link>
<Nav.Link as={NavLink} to="/about" style={({ isActive }) => (isActive ?
↪  activeStyle : undefined)}>About</Nav.Link>
```

- Note to include Link from react-router in Navbar
- Now you should be able to switch between home & about pages
- When we click on know more we like to see the details of a listing
- Create a listing details component.

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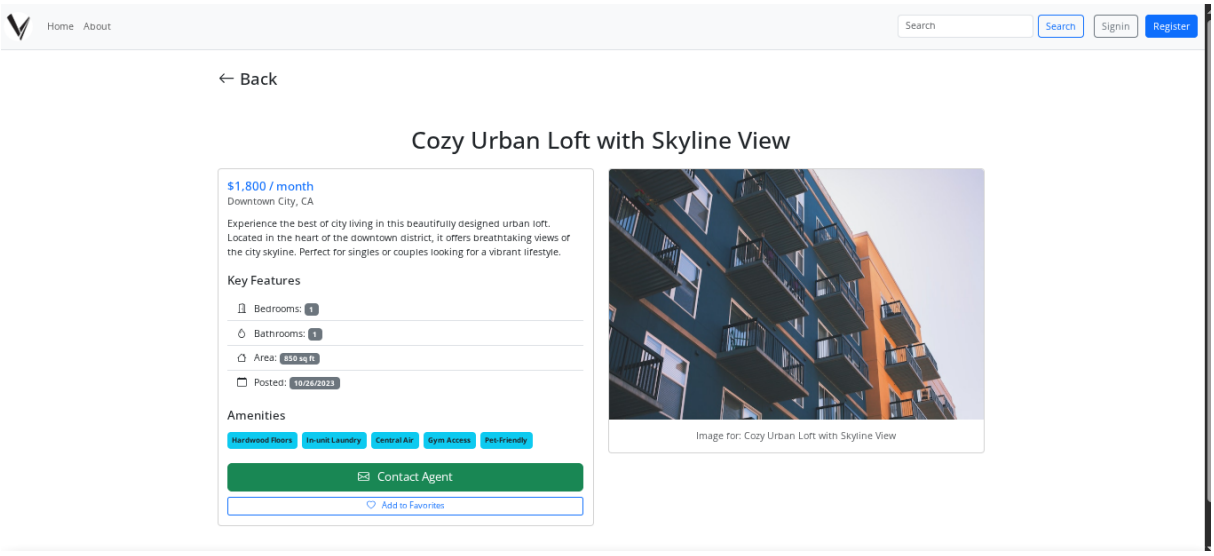
LLD of Listing Details

HLD of Listing Details



- clicking on know more will route to that page

1. add route in the app routes

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```
<Route path="/listings/:listingId" element={<ListingDetails />} />
```

2. Create Component

```
import React, { useState } from "react";

export default function ListingDetails(){
    const { listingId } = useParams();
    const [listing, setListing] = useState(null);
    const [loading, setLoading] = useState(true);
    const [error, setError] = useState(null);

    return (
        <>
        <h1>{listingId}</h1>
        </>
    )
}
```

3. Link to the know more

```
<Link to={`/listings/${listing.id}`} >
  {linkText}
</Link>
```

- The above will display the id of the listing
- React Router's useEffect hook is often used for side effects, such as fetching data, when a component mounts or when specific dependencies change.
- Also we will modify our data source to be from a `json-server` based on `db.json` as below

```
{
    "locations": [
        {
            "id": 0,
            "name": "Acme Fresh Start Housing",
            "city": "Chicago",
            "state": "IL",
            "photo": "https://angular.dev/assets/images/tutorials/common/bernard-hermant-CLKGGwIBTaY-unsplash.jpg",
            "availableUnits": 4,
            "wifi": true,
            "laundry": true
        },
        {
            "id": 1,
            "name": "A113 Transitional Housing",
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            "availableUnits": 0,
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```


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- Further instructions will be added in the final document

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